A5 · Errors on Purpose

CS425 Computer Networks · instructor led · every command runs on Onyx (`ssh onyx`)

TURN THIS IN ON PAPER
before you leave the room
graded pass/fail

Name _____ Date _____

Every round: **predict** first, in ink, then **run** the command with the projector, then **check**. A wrong prediction costs nothing. A prediction written afterwards is not a prediction.

ROUND 1 · IS ANYBODY OUT THERE? `ping · curl`

QUESTION	ANSWER
Predict: will `ping -c 3 8.8.8.8` get replies?	
What did `ping 8.8.8.8` report?	
Did the gateway `132.178.227.1` answer? What did `curl` print?	

Q1. So what is actually broken? What do you now know about how campus treats ICMP echo, compared with TCP on port 443?

ROUND 2 · TRACEROUTE BY HAND `ping -c 1 -t TTL 8.8.8.8`

TTL	PREDICT: WHAT COMES BACK, AND FROM WHOM	WHAT ACTUALLY CAME BACK
`-t 1`		
`-t 2`		
`-t 3`		

Q2. Who sent the `Time to live exceeded`, and how do you know from the output alone? Nothing came back at `-t 2` and `-t 3`. Does that mean there is no router at hop 2 or hop 3?

ROUND 3 · HOW DO YOU KNOW YOU ARRIVED? `UDP to 198.60.22.13 port 33434`

QUESTION	ANSWER
Predict: a reply, a 3 second timeout, or something else?	
What did Python print? And for `127.0.0.1`?	

Q3. UDP has no connection, so what was refused, and who said so? Which ICMP message came back, and how did it become an error in a Python program?

ROUND 4 · SAME PATH, THREE WAYS `mtr -n [-u | -T -P 443] -r -c 5 mirrors.xmission.com`

PROBE	PREDICT: HOW FAR WILL THE TRACE GET?	LAST HOP THAT ANSWERED	HOP 2'S ADDRESS, IF ANY
ICMP echo (default)			
UDP `-u`			
TCP `-T -P 443`			

Q4. Which hop do you think is throwing away your pings? What is your evidence? What kind of address does hop 2 have (think back to A4, round 1)?

Q5. Hops 6 to 9 lose 100% of the probes, yet hop 13 loses none. How can a router drop every probe and still forward all of your traffic?

Exit. A friend runs `ping 8.8.8.8` on Onyx, sees 100% packet loss, and says Onyx is off the Internet. Give them two pieces of evidence from today that say otherwise.

Graded pass/fail on something in every box. Hand this in before you leave.